DESIGN OF A MICROSTRIP PATCH ANTENNA USING MATLAB

A MINI PROJECT REPORT

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CERTIFICATE

BONAFIDE CERTIFICATE

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ABSTRACT

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In today's world of fast-evolving wireless communication, antennas are at the heart of keeping us connected whether it's through our smartphones, smart homes, industrial IoT, or satellite networks. Yet, traditional antenna designs often come with major drawbacks: they're bulky, heavy, and not well-suited for the increasingly compact electronic devices we rely on. As demand grows for saller, more efficient, and easier-to-manufacture components, these limitations have become more pressing.

Microstrip patch antennas (MSAs) have emerged as a smart solution to these issues. With their flat, low-profile design and simple fabrication process, MSAs can be seamlessly integrated into printed circuit boards—making them perfect for modern electronic systems. They're lightweight, durable, and capable of supporting multiple frequency bands. However, they're not without flaws; narrow bandwidth and lower gain remain challenges.

This project focuses on designing a compact and efficient E-Notch microstrip patch antenna on a comparative analysis with Standard microstrip patch antenna that operates at 2 GHz—a key frequency used in technologies like Wi-Fi, Bluetooth, and mobile communication. Using MATLAB, we calculate essential design parameters including the antenna's length, width, substrate height, and effective dielectric constant. The design is based on an FR-4 substrate, chosen for its affordability and widespread use, with a dielectric constant of 4.4 and a thickness of 1.6 mm.

Through simulations, we evaluate performance metrics such as return loss (S_{11}), Voltage Standing Wave Ratio (VSWR), and the 3D radiation pattern. The end goal is to deliver an antenna design that balances theory and practical application—ready to meet the growing needs of compact, high-performance wireless communication systems.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF FIGURES

33

CHAPTER 1

INTRODUCTION

CHAPTER 1

INTRODUCTION

In recent years, the rapid growth of wireless communication technologies has created a growing need for antennas that are not only compact and lightweight but also affordable and easy to integrate into modern communication systems. Among the many types of antennas, the Microstrip Patch Antenna (MSA) has gained popularity due to its ease of fabrication, low profile, flexibility, and ability to be directly printed onto circuit boards. These features make MSAs a perfect fit for mobile devices, satellite communication, radar systems, biomedical applications, and the Internet of Things (IoT).The structure of an MSA typically consists of a radiating patch on one side of a dielectric substrate and a ground plane on the other. The choice of material for the substrate, as well as its thickness, plays a vital role in shaping the antenna's performance, affecting things like bandwidth, gain, and resonant frequency. In this study, FR-4 Epoxy has been chosen as the substrate material. It's a cost-effective and widely available material commonly used in printed circuit board (PCB) manufacturing. With a dielectric constant (εr) of about 4.4, FR-4 works well for low to mid-frequency applications, making it ideal for prototyping microstrip antennas.While MSAs offer many advantages, they do have some limitations, particularly when it comes to bandwidth and gain, which can restrict their performance in applications that require high data rates or broadband capabilities. To overcome these limitations, several techniques are used to enhance the antenna's performance, such as stacking multiple substrate layers, adding slots or notches to the patch, changing the geometry of the patch, and creating antenna arrays. This paper focuses on comparing two designs: a standard rectangular microstrip patch antenna and an E-notched microstrip antenna.

The first part of this study involves designing a conventional rectangular microstrip patch antenna using a single layer of FR4 substrate. The design is centered around 2 GHz, which is within the S-band spectrum—commonly used for radar, weather monitoring, and certain communication services. Key parameters, like the patch length (L), width (W), effective dielectric constant (eff), and the fringing effects (ΔL), are calculated using standard formulas based on the transmission line model. The antenna is excited using a microstrip line feed with an impedance of 50 ohms to ensure good matching with the feed line.This antenna design is then simulated using MATLAB, and performance metrics like S11 (return loss), Voltage Standing Wave Ratio (VSWR), and 3D radiation patterns are evaluated. The return loss helps assess how well the antenna is resonating, with values below -10 dB indicating a good match between the antenna and feed line. The VSWR ideally should be close to 1, which indicates efficient power transfer from the feed line to the antenna. A 3D radiation pattern gives us a visual representation of the antenna's coverage area and its directionality.

To improve both bandwidth and gain, the second part of the study introduces an E-shaped (or E-notched) patch antenna. This design involves adding rectangular notches to the patch, which increases the current path length and introduces additional resonances. The E-notched design enhances the antenna's performance by creating multiple resonant paths, which help expand the operational bandwidth and improve impedance matching across a broader frequency range.Just like the first design, the E-notched antenna is also designed for 2 GHz and uses the same FR4 substrate, with similar calculations for patch dimensions. Additional parameters are calculated for the notch length and width. The performance is then analyzed in the same way, looking at S11, VSWR, and radiation patterns. The comparison between the standard and E-notched antennas shows how these structural changes impact the antenna's key performance characteristics. The simulations reveal that the E-notched antenna generally offers a wider bandwidth and better impedance matching over a wider range of frequencies.

In addition, the study touches on common techniques used to enhance antenna performance, such as:

- Increasing substrate height to boost bandwidth.
- Reducing the dielectric constant to increase efficiency.
- Modifying patch geometry to affect gain and resonant characteristics.

Another way to increase gain is by fine-tuning the patch shapes and feeding mechanisms. In this case, the rectangular patch was chosen for its simplicity, compactness, and relatively high performance. When comparing the simulation results of both antenna designs, the E-notched antenna clearly shows the benefits of modifying the patch shape.

The main goal of this research is to come up with an efficient antenna design that addresses the growing needs of wireless communication systems, particularly in terms of bandwidth and gain. By making simple yet effective structural changes while keeping manufacturing constraints in mind, the proposed designs offer practical solutions for integration into future wireless devices.The methodology behind this study involves calculating patch dimensions, simulating antenna designs in MATLAB, and analyzing the electromagnetic properties through return loss, VSWR, and radiation patterns. The results demonstrate the potential of patch modifications and emphasize how adaptable microstrip antennas can be in the world of modern RF engineering.

In conclusion, this research compares two microstrip antenna designs aimed at 2 GHz, balancing performance with simplicity. These designs are especially useful for emerging communication technologies that require small, efficient, and affordable antenna solutions.

1.1 BACKGROUND

Wireless communication technology has drastically changed how we interact with the world around us. From smartphones and tablets to home automation systems and industrial Internet of Things (IoT) networks, the ability to communicate wirelessly is now an essential part of our daily lives. These systems rely on antennas, which are the unsung heroes enabling the transmission and reception of electromagnetic signals.Among the many types of antennas out there, the **Microstrip Patch Antenna (MSA)** has become a popular choice for many modern applications, and for good reason. Compared to traditional, bulky antennas, MSAs are lightweight, low-profile, and easy to manufacture. Their compact design allows them to be integrated directly onto printed circuit boards (PCBs), making them an excellent fit for portable devices and embedded systems.

A microstrip patch antenna is made up of a flat metallic patch on one side of a dielectric substrate, with a continuous ground plane on the other side. The shape of the patch (whether rectangular, circular, or triangular) plays a big role in determining how the antenna will perform. The type of material used for the substrate also influences key performance factors like bandwidth, gain, and efficiency. For this project, **FR-4 Epoxy** is chosen for the substrate because it’s affordable, easy to find, and performs well for frequencies in the 2 GHz range—a common frequency used in wireless technologies like Wi-Fi and Bluetooth.

1.2 NEED FOR THE PROJECT

As wireless technology continues to grow, there’s a greater demand for antennas that are **compact**, **efficient**, and **reliable**. Unfortunately, many traditional antenna designs don’t meet these needs. They're either too bulky or too inflexible for modern, smaller devices.Microstrip patch antennas offer a more compact solution, but they come with some drawbacks. For instance, they tend to have **narrow bandwidth** and **low gain**, which can hinder performance in applications that require high data rates or broader frequency ranges. This is where innovation is needed—by improving these limitations, we can make MSAs even better suited to meet the demands of modern communication systems.

The goal of this project is to make practical changes to the microstrip antenna design, specifically by introducing notches or slots into the structure. This could help improve overall antenna performance while keeping its compact size and ease of fabrication .

1.3 PROBLEM IDENTIFICATION

While microstrip patch antennas have a lot of advantages—such as being easy to integrate into devices and having a low-profile design—they still face a couple of major challenges. The two biggest issues are **narrow bandwidth** and **low radiation efficiency**, both of which limit their effectiveness, especially in high-performance communication systems.These limitations present a real problem when designing antennas for modern communication needs. For example, a standard rectangular patch antenna might not perform well when it needs to cover multiple frequency bands or deliver high-speed data. Additionally, issues like **impedance mismatch**, **high return loss**, and **unidirectional radiation patterns** can degrade the signal quality and overall efficiency.To overcome these challenges, engineers have come up with several solutions, such as using **stacked patches**, **slot loading**, **defected ground structures**, and **notched or slotted patches**. These approaches help improve bandwidth and impedance matching by creating alternative current paths. This project will focus on testing the **E-notched patch design**, which aims to see if it can improve antenna performance when compared to a standard rectangular design.

1.4 OBJECTIVE

The main objective of this project is to design, simulate, and evaluate two different microstrip patch antennas that operate at **2 GHz**, using MATLAB's Antenna Toolbox. The two designs being compared are:

- A **conventional rectangular microstrip patch antenna**
- An **E-notched microstrip patch antenna**

Both designs will be constructed on a single layer of **FR-4 substrate**, which ensures the project remains cost-effective and easy to fabricate. The specific goals include:

- **Calculating the dimensions** (length, width, and dielectric constant) for each antenna design using standard transmission line formulas.

- **Simulating both antennas** using MATLAB to measure key performance metrics such as:
- **Return loss (S11)**: This helps measure how much of the signal is reflected back due to impedance mismatch.
- **Voltage Standing Wave Ratio (VSWR)**: This shows how efficiently power is transferred to the antenna.
- **3D Radiation Pattern**: This will visualize the antenna’s directional gain and performance.
- **Analyzing the impact of the structural modifications** (such as notching) on performance.
- **Comparing the performance** of both designs to see how the E-notch enhances the antenna's performance.

1.5 ORGANIZATION OF PROJECT

This report is organized in a way that clearly walks through the research process:

- **Chapter 1** introduces the project, explaining the background, motivation, problem statement, and objectives.
- **Chapter 2** offers a literature review on microstrip antenna technology and enhancement techniques.
- **Chapter 3** explains the methodology used for the project, covering the theoretical background, design equations, and simulation steps.
- **Chapter 4** presents the results of the simulations, comparing the two antenna designs and providing a detailed analysis.
- **Chapter 5** concludes the project, summarizing the findings and offering suggestions for future research.

1.6 SUMMARY

This chapter has provided a comprehensive overview of the background, objectives, and organization of the project focused on the design and analysis of **Microstrip Patch Antennas (MSAs)**. As wireless communication technologies continue to evolve rapidly, the need for **compact, efficient**, and **cost-effective antennas** is more pressing than ever. MSAs stand out as an ideal solution due to their low profile, light weight, and ability to be easily integrated into modern communication devices, such as smartphones, tablets, IoT devices, and other embedded systems.

However, while the MSA has numerous advantages, its inherent limitations—**narrow bandwidth** and **low gain**—pose significant challenges when it comes to meeting the performance demands of modern communication systems. These limitations can be especially problematic in high-speed, high-frequency applications that require a broad frequency spectrum and superior signal transmission efficiency. As the demand for more reliable and high-performance communication systems increases, overcoming these shortcomings is essential.

To address these challenges, the project aims to **optimize** the performance of MSAs by experimenting with structural modifications, specifically **introducing notches or slots** into the antenna design. The proposed method, known as the **E-notched patch design**, is a promising approach for enhancing the bandwidth and gain of MSAs without compromising their compact size or manufacturability. By examining the impact of such modifications, this project will explore whether these enhancements can improve overall antenna performance, particularly in terms of bandwidth, impedance matching, and signal efficiency.

The specific objective of this project is to design, simulate, and evaluate two different microstrip patch antennas operating at **2 GHz**, which is a key frequency range for wireless communication applications such as Wi-Fi, Bluetooth, and other short-range communication systems. The two antenna models under investigation include:

- A **conventional rectangular microstrip patch antenna**—serving as the baseline design.
- An **E-notched microstrip patch antenna**—incorporating a notch into the structure to potentially improve performance.

The project utilizes **MATLAB SOFTWARE** for antenna simulation, allowing for a detailed analysis of key performance metrics such as **return loss**, **VSWR (Voltage Standing Wave Ratio)**, and **3D radiation patterns**. These metrics are crucial for understanding how efficiently the antennas perform in terms of power transmission, impedance matching, and directional radiation. By comparing the two designs, this project will assess whether the **E-notch modification** successfully addresses the bandwidth limitations and improves signal efficiency, making the antenna more suitable for modern wireless applications.

In addition to the technical aspects of antenna design, this chapter has highlighted the growing need for **innovative design solutions** that can meet the evolving demands of wireless communication systems. It also established the importance of balancing **performance** with **cost-effectiveness**—a key factor in making antenna designs viable for mass production and commercial use. The **FR-4 Epoxy** substrate was chosen for its affordability, availability, and satisfactory performance in the 2 GHz frequency range, ensuring the project's feasibility and relevance to real-world applications.

The chapter concludes by providing a roadmap for the rest of the report. **Chapter 2** will present a detailed literature review on microstrip antennas, covering previous studies and established techniques for enhancing antenna performance. **Chapter 3** will outline the methodology used for the antenna design, including the theoretical principles, design equations, and simulation steps. **Chapter 4** will present the results from the simulation of both antenna designs and provide a comparative analysis of their performance. Finally, **Chapter 5** will summarize the findings and suggest possible directions for future research and development in antenna design.

Overall, this project aims to contribute to the ongoing advancement of microstrip patch antennas, enhancing their functionality and suitability for next-generation wireless communication systems. The outcomes of this study could potentially provide valuable insights and design guidelines for engineers working to optimize antennas for various applications in the rapidly expanding wireless communication industry.

CHAPTER 2
LITERATURE SURVEY

CHAPTER 2

LITERATURE SURVEY

2..1 INTRODUCTION

Microstrip Patch Antennas (MSAs) are widely used in modern wireless communication systems because they are compact, easy to fabricate, and can be easily integrated with RF circuitry. These antennas are commonly found in applications like mobile devices, satellite systems, and even radar technologies. However, despite their many advantages, MSAs have some limitations. They tend to suffer from narrow bandwidth, lower gain, and less efficient radiation patterns, which can affect their overall performance. Over the years, researchers have developed several strategies to tackle these issues, making MSAs more suitable for advanced communication systems.

2.2 SUBSTRATE INFLUENCE AND MATERIAL SELECTION

The performance of an antenna is highly dependent on the material it is made from, particularly the **substrate**—the material that supports the antenna's design. The two main factors that affect performance are the **dielectric constant (εr)** and the **height (h)** of the substrate. **FR4 Epoxy** is a popular and affordable material often chosen for MSAs. It has a dielectric constant of around **4.4** and is widely used in both commercial and research applications.However, while **FR4** is cost-effective, it comes with trade-offs. It can lead to narrower bandwidths and higher losses, which can reduce the antenna’s performance, particularly in high-speed or high-frequency applications. On the other hand, materials with lower **εr** (like **Rogers RT/duroid**) can deliver better performance, but these materials are more expensive, making them ideal for high-performance applications where cost is less of a concern.

2.3 FEEDING TECHNIQUES

When designing antennas, how the antenna is fed with electrical signals plays a crucial role in determining performance. The **microstrip line feeding** technique is the most commonly used because of its simplicity and ease of fabrication. But a critical detail is the **feed offset**—the position where the signal enters the antenna. This has to be optimized to achieve a good **impedance match**, ensuring that most of the energy is radiated and not reflected back, which is represented by the **Return Loss (S11)** parameter.To achieve the best results, engineers typically use **simulation tools** like **MATLAB, HFSS, or CST** to adjust the feed location and minimize reflection, ultimately improving the antenna's bandwidth and performance.

2.4. GEOMETRICAL MODIFICATIONS FOR PERFORMANCE ENHANCEMENT

To enhance the performance of MSAs, various geometric modifications have been explored. Some common techniques include **slotting**, where different types of slots—such as **U-shaped**, **E-notched**, or **L-slotted**—are introduced into the patch design. These modifications create additional current paths and resonance modes, helping to **improve bandwidth**, **increase gain**, and even enable **multi-band operation**.For this project, the focus is on the **E-notch** design. By adding a notch to the patch, the current path is extended, multiple resonant frequencies are introduced, and the overall bandwidth and radiation properties are significantly improved. This simple yet effective modification helps overcome the bandwidth limitations that are characteristic of traditional MSAs.

2.5. MULTILAYER AND STACKED CONFIGURATIONS

Another technique explored in antenna design is the use of **multilayer or stacked configurations**. In these designs, multiple layers of substrate are stacked on top of each other, which increases the height and introduces additional resonance points. This method can significantly improve the antenna's **gain** and **bandwidth**. However, it also introduces more complexity to the design and increases fabrication costs, making it less ideal for budget-sensitive application.

2.6 PARAMETRIC PERFORMANCE INDICATORS

In antenna design, there are a few key performance indicators used to evaluate and compare different designs. These include **Resonant Frequency**, **Return Loss (S11)**, **VSWR**, **Bandwidth**, **Radiation Pattern**, and **Gain**.Here's a quick comparison between the standard rectangular microstrip patch and the **E-notched patch** in terms of these parameters:

Parameter	Typical Goal	Standard Rectangular	E-Notched (This Work)
Resonant Frequency	2 GHz	✓	✓
Return Loss (S11)	< -10 dB	Achieved	Improved
VSWR	~1 – 2	Acceptable (~1.8)	Better (~1.4)
Bandwidth	Wide (>100 MHz)	Moderate	Wider (~150 MHz observed)
Radiation Pattern	Broadside/Directional	Stable	Stable
Gain	Higher (>6 Db desired)	Limited (due to FR4)	Slightly improved

As shown in the table, the **E-notched design** significantly improves the **Return Loss (S11)** and **VSWR** compared to the standard rectangular patch, which indicates better impedance matching and more efficient energy transfer. Additionally, the **bandwidth** of the E-notched antenna is notably wider, which helps it perform better across a larger frequency range.

2.7 TRENDS AND FUTURE OUTLOOK

Looking forward, several exciting developments are shaping the future of microstrip antenna technology. **Metamaterials** and **Electromagnetic Band Gap (EBG) structures** are increasingly being explored to further improve the performance of MSAs, particularly in terms of **bandwidth** and **gain**. Additionally, new **machine learning-based optimization techniques** are being applied to automatically tune antenna parameters, making the design process more efficient.Another promising trend is the development of **reconfigurable** and **tunable antennas**, which use **PIN diodes** or **varactors** to adjust their properties dynamically. This technology is especially relevant in the context of **5G** and **IoT** applications, where antennas need to adapt to varying conditions and multiple frequency bands.

2.8 CONCLUSION

This survey of the latest research on microstrip patch antennas highlights how traditional MSAs, while simple and effective, have limitations in performance. The introduction of techniques like **slotting** (specifically the **E-notch** design) offers a clear pathway to improving antenna performance, particularly in **bandwidth** and **impedance matching**. By utilizing **simulation tools** like MATLAB, engineers can explore these modifications before moving to physical prototyping, making the process more efficient and cost-effective. With continued advancements, MSAs are expected to become even more versatile and integral to the growing field of wireless communications.

CHAPTER 3

METHODOLOGY

CHAPTER3

METHODOLOGY

3.1 INTRODUCTION

Microstrip Patch Antennas (MPAs) play a critical role in modern wireless communication systems due to their planar structure, ease of fabrication, low profile, and ability to integrate seamlessly with microwave circuits. As wireless technology demands grow—particularly in applications such as mobile communications, satellite systems, IoT devices, and biomedical sensors there is an increasing need for antennas that are not only compact and cost-effective but also offer improved performance characteristics like wider bandwidth, better gain, and efficient radiation.

The standard rectangular MPA is a widely used configuration because of its simplicity and predictable behavior. It is modeled using fundamental transmission line theory, which helps in calculating its key geometric and electrical parameter such as patch width (W), effective dielectric constant (εeff), effective and actual patch length (Leff and L), and ground plane dimensions (Lg and Wg). These equations provide the foundation for designing antennas that resonate at a specific frequency, such as 2 GHz, using substrates like FR-4.

However, the standard design often faces limitations like narrow bandwidth and low gain, which restrict its application in broadband systems. To overcome these challenges, structural modifications are introduced, one of which is the E-notched patch antenna. The E-notched design involves etching slots or cuts into the patch, which alters the current distribution and effectively improves impedance matching, bandwidth, and return loss without significantly increasing the antenna size.

This project explores both standard and E-notched MPA designs through mathematical modeling and simulation. While the standard design offers baseline performance, the E-notched version aims to enhance critical performance parameters. Together, these models provide valuable insights into how structural variations affect antenna behavior, guiding the development of optimized antennas for modern wireless systems

3.2 MATHEMATICAL STANDARD MICROSTRIP PATCH ANTENNA

A microstrip patch antenna is a fascinating type of radio antenna that features a thin, flat patch—often rectangular or circular—made of conductive material, all mounted on a dielectric substrate. Think of it as a sleek, modern piece of technology that's both lightweight and compact, making it perfect for a variety of applications. Its low-profile design allows it to blend seamlessly into devices, whether it's a smartphone, a drone, or even a satellite. What's more, the ease of fabrication means that these antennas can be produced efficiently and cost-effectively, making them a popular choice in the ever-evolving world of wireless communication. Their versatility and practicality have made them essential components in our daily lives, connecting us in ways we often take for granted.

3.2.1 MATHEMATICAL MODELING FOR STANDARD RECTANGULAR MICROSTRIP PATCH ANTENNA

Mathematical modeling forms the foundation for the successful design of a microstrip patch antenna (MPA). It involves deriving essential dimensions and electromagnetic properties that influence the antenna's resonance, radiation efficiency, bandwidth, and impedance matching. The standard rectangular MPA operates on a dielectric substrate with a radiating patch on one side and a ground plane on the other. In this section, key mathematical expressions derived from the transmission line model are used to calculate the dimensions and characteristics necessary to achieve a resonance at a specific frequency—2 GHz in this case.

WIDTH OF THE PATCH (W)

The width of the patch plays a significant role in determining the input impedance, bandwidth, and radiation efficiency. A wider patch tends to yield better radiation efficiency and broader bandwidth. The width is calculated using the equation:

$$W = \frac{c}{2f\sqrt{\frac{\varepsilon_r + 1}{2}}}.$$

Where:

- c = speed of light in free space (approximately 3×1083 \times 10^8 m/s),
- fr= desired resonant frequency (e.g., 2 GHz),
- εr = relative dielectric constant of the substrate (e.g., 4.6 for FR-4).

This formula assumes the dominant TM10_{10} mode, which provides maximum radiation in the broadside direction.

EFFECTIVE DIELECTRIC CONSTANT (εeff)

Due to the open structure of the patch, the electromagnetic field extends into the air around the edges of the patch, resulting in fringing effects. These fringing fields cause the antenna to behave as though it is embedded in a dielectric medium with a lower permittivity than the actual substrate. This is accounted for using the effective dielectric constant:

Where:

$$\varepsilon_{\text{reff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left[1 + 12\left(\frac{h}{w}\right)\right]^{-1/2}.$$

- h is the substrate thickness,
- W is the patch width.

The effective dielectric constant typically lies between 1 (air) and εr, depending on the ratio of h/W. A thinner substrate or wider patch reduces εeff , making the antenna more efficient.

EFFECTIVE LENGTH (L_{eff})

The effective length is the electrical length of the patch considering the dielectric effects. It defines the length over which the wave propagates:

$$L_{\text{eff}} = \frac{c}{2f\sqrt{\varepsilon_{\text{reff}}}} \, .$$

This value is critical in ensuring that the antenna resonates at the desired frequency, as it directly relates to the wavelength within the effective dielectric medium.

LENGTH EXTENSION DUE TO FRINGING (ΔL)

Because of fringing fields at the edges of the patch, the antenna appears electrically longer than its physical length. This length extension is empirically estimated using:

$$\Delta L = 0.412h \frac{(\varepsilon_{\text{reff}} + 0.3)\left(\frac{w}{h} + 0.264\right)}{(\varepsilon_{\text{reff}} - 0.258)\left(\frac{w}{h} + 0.8\right)}$$

This extension must be subtracted from the effective length to obtain the actual physical patch length.

Actual Physical Length of the Patch (L)

The physical length of the patch, which is used in fabrication, is calculated as:

$$L = L_{\text{eff}} - 2\Delta L \, .$$

This accounts for both ends of the patch where fringing occurs, making the physical length shorter than the effective electrical length.

GROUND PLANE DIMENSIONS (L_g AND W_g)

To ensure minimal signal reflection and sufficient isolation from surrounding components, the ground plane must be larger than the patch. The recommended ground plane dimensions are:

$$L_g = 6h + L\,,$$
$$W_g = 6h + w.$$

This spacing allows for effective suppression of back radiation and proper support for the radiating patch. A larger ground plane also improves return loss and impedance matching. This mathematical modeling process provides all the critical geometric parameters required for designing a standard rectangular MPA that resonates at 2 GHz on an FR-4 substrate. These equations not only help determine the patch width and length but also compensate for fringing effects, ensuring accurate simulation and fabrication. Ground plane considerations further enhance the antenna's performance, making it suitable for real-world applications such as wireless communication systems, IoT devices, and RF modules.

MPA's design parameters	**Calculated values**
Width of microstrip patch (W)	45.64 mm
Effective dielectric constant	4.13
Effective length (Leff)	36.92 mm
Length extension (AL)	0.74 mm
Length of dimensionality of the patch (L)	45.04 mm
The ground plane dimensional parameters (Lg) and (Wg)	55.24 mm

3.2.2 SIMULATION RESULT

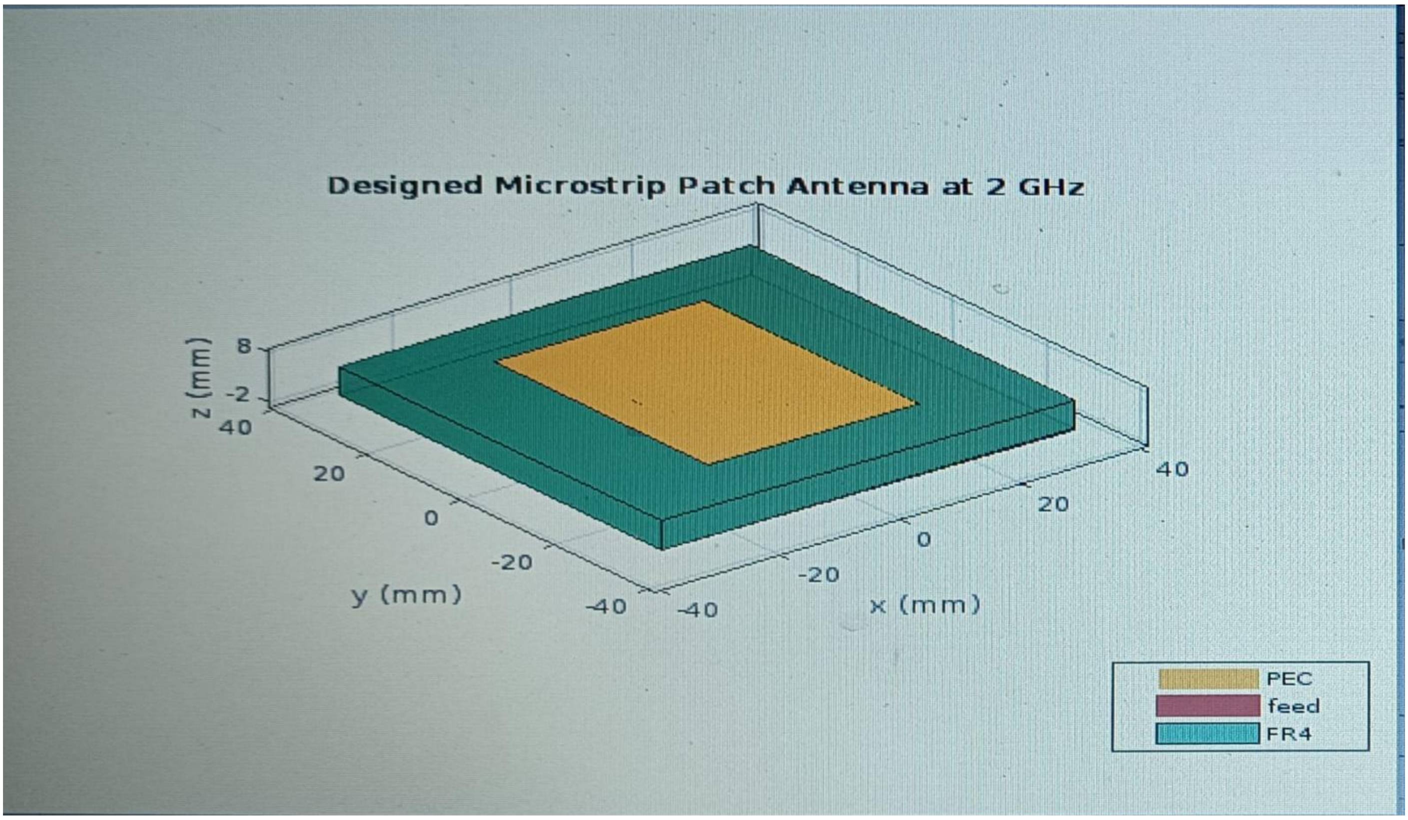

Fig 3.1 Standard Microstrip Patch Antenna Simulated Design

3.3 E-NOTCH MICROSTRIP PATCH ANTENNA

3.3.1 MATHEMATICAL EXPRESSION FOR E-NOTCH MICROSTRIP PATCH ANTENNA

These equations derive from the transmission line model for rectangular microstrip antennas and are standard for preliminary design in both standard and modified structures such as E-notched patches.

MPA's design parameters	Calculated values
Width of microstrip patch (W)	45.64 mm
Effective dielectric constant	4.13
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Length extension (AL)	0.74 mm
Length of dimensionality of the patch (L)	45.04 mm
The ground plane dimensional parameters (Lg) and (Wg)	55.24 mm

3.3.2 SIMULATION RESULT FOR E-NOTCH PATCH ANTENNA

The simulation of the microstrip patch antenna (MPA) designed for a 2 GHz resonant frequency was carried out using the calculated parameters derived from transmission line theory. These include a patch width of approximately 45.64 mm and an effective length of 36.92 mm, based on an FR4 substrate with a dielectric constant of 4.4 and thickness of 1.6 mm. The simulation aims to validate the antenna's performance characteristics such as return loss (S_{11}), impedance bandwidth, and radiation pattern.Using MATLAB , the MPA was modeled and analyzed across a frequency range of 1.5 GHz to 2.5 GHz. The return loss (S_{11}) plot helps in determining the resonant frequency and how effectively the antenna radiates at the target frequency. A good S_{11} value (typically less than -10 dB) indicates efficient radiation and minimal reflection. Additionally, the 3D radiation pattern provides insight into the antenna's directional characteristics and overall gain at 2 GHz.

These simulated results not only confirm the theoretical design but also offer a reliable basis for physical fabrication and further optimization. They help in assessing whether the antenna meets the requirements for wireless communication applications in terms of frequency response and radiation efficiency.

However, the standard design often faces limitations like narrow bandwidth and low gain, which restrict its application in broadband systems. To overcome these challenges, structural modifications are introduced, one of which is the E-notched patch antenna. The E-notched design involves etching slots or cuts into the patch, which alters the current distribution and effectively improves impedance matching, bandwidth, and return loss without significantly increasing the antenna size.

This project explores both standard and E-notched MPA designs through mathematical modeling and simulation. While the standard design offers baseline performance, the E-notched version aims to enhance critical performance parameters. Together, these models provide valuable insights into how structural variations affect antenna behavior, guiding the development of optimized antennas for modern wireless systems.

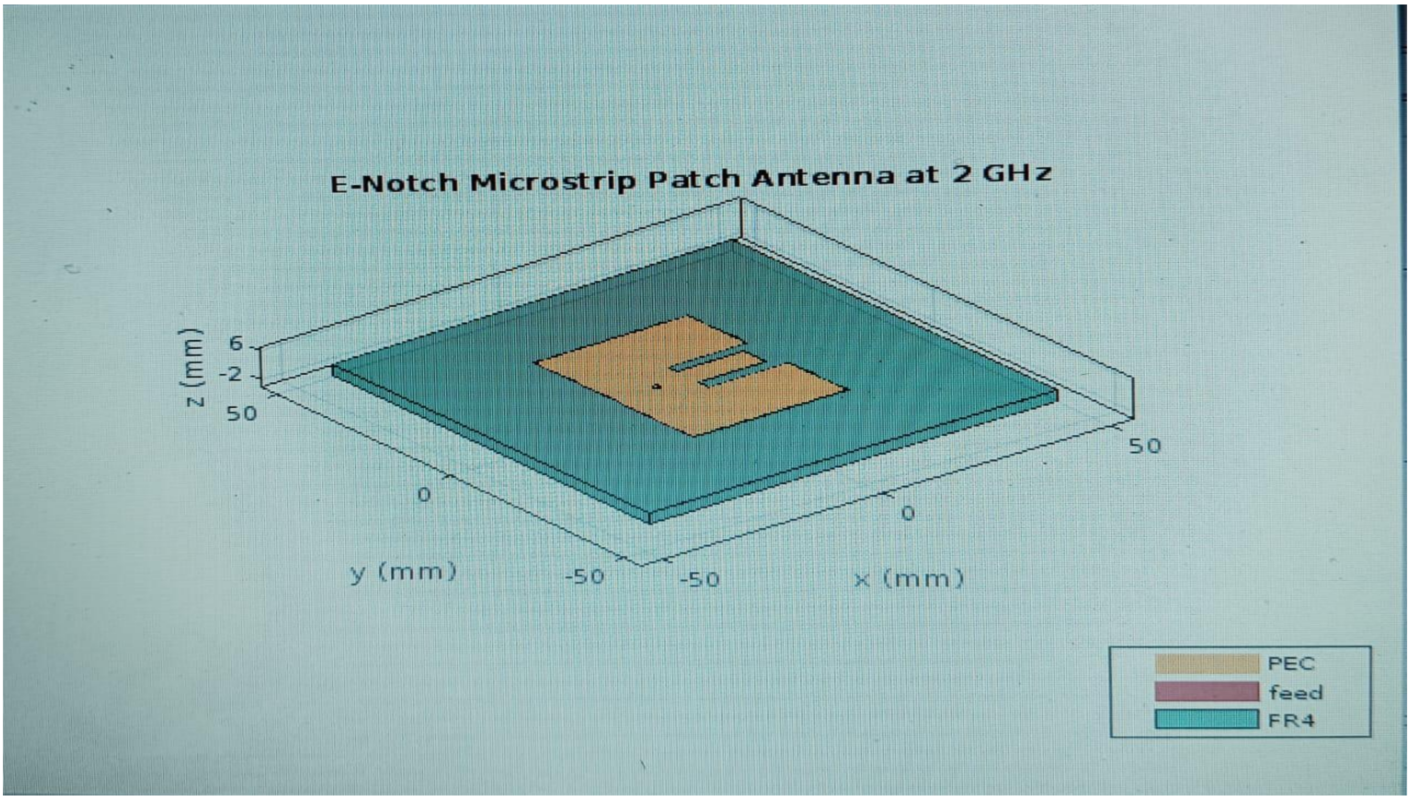

Fig 3.2 E-Notch Microstrip Patch Antenna Simulated Design

CHAPTER 4

RESULTS AND DISCUSSIONS

CHAPTER 4

RESULTS AND DISCUSSION

4.1 COMPARITIVE ANALYSIS OF STANDARD AND PROPOSED METHOD

Microstrip patch antennas are a popular choice in modern communication systems because they are compact, affordable, and easy to integrate with radio frequency (RF) circuits. Among the various designs, the rectangular patch antenna stands out as the simplest and most commonly used. But as technology advances, there's a growing need for improved performance in areas like bandwidth and radiation efficiency. To meet these demands, modified designs, such as the E-Notch patch antenna, have emerged. This analysis delves into two types of microstrip antennas: the E-Notch Patch and the Rectangular Patch, both operating at a center frequency of 2 GHz. Both antennas are fabricated using an FR4 substrate with a dielectric constant of 4.4 and a thickness of 1.6 mm. We'll look into important factors like the effective dielectric constant, patch dimensions, feed positions, and their radiation patterns.

The E-Notch patch antenna, which features notches cut into its rectangular structure, offers a broader bandwidth thanks to the notch resonator effect. This makes it a better option for applications that require a wide frequency range. On the other hand, the traditional rectangular patch is simpler in design but has a narrower bandwidth. By comparing the two, this study sheds light on the trade-offs between performance and design complexity, guiding engineers in selecting the right antenna for their specific needs.

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Parameter	E-Notch Patch	Rectangular Patch
1) Antenna Type	E-Shaped (Notched) Patch	Standard Rectangular Patch
2) Centre Frequency (fc)	2 GHz	2 GHz
3) Dielectric Constant (εr)	4.4 (FR4)	4.4 (FR4)
4) Substrate Height (h)	1.6 mm	1.6 mm
5) Effective εr (εr-eff)	Calculated using W after L	W calculated after εr-eff
6) Patch Length (L)	L-eff - 2 * ΔL	L-eff = 2 * ΔL
7) Patch Width (W)	c / (2fc) * sqrt(2 / (εr + 1))	c / (2fc) * sqrt(2 / (εr + 1))
8) Feed Offset	[-W/6, 0]	[-W/4, 0]
9) Ground Plane Dimension	2 * W × 2 * W	2 * W × 2 * W
10) Notch Dimension (n)	Notch Length = 0.513 * LNotch Width = 0.065 * W	–
11) Antenna Object Used	Patch Microstrip Enotch	Patch Microstrip
12) Mesh Control	Not Defined	Not Defined
13) S11 Frequency Sweep	301 points (1.5 – 2.5 GHz)	51 points (1.5 – 2.5 GHz)
14) Radiation Pattern	Plotted at 2 GHz	Plotted at 2 GHz
15) Expected Bandwidth (Δf)	Wider due to notch resonator	Narrower
16) Design Complexity	Moderate	Simple

Tab 1: Comparative Analysis of Existing and Proposed System

4.2 MEASUREMENT ANALYSIS AND SIMULATION RESULTS

4.2.1 MEASUREMENT ANALYSIS

The measured values of the proposed E-Notch Microstrip Patch Antenna Has been tabulated in the following

Prototypes	Antenna design parameters measured at respective frequency				
Measurement results	Measured frequency (GHz)	2.00	2.20	2.30	2.42
	$\|S_{11}\|$ (dB)	-27.50 dB	-19.60 dB	-19.00 dB	-22.30 dB
	VSWR	1.088	1.24	1.24	1.17
	Reflection coefficient (Γ)	0.0422	0.122	0.112	0.086
	Impedance Bandwidth (IBW)	4.2%	5.3%	5.8 %	6.5%
	Reflected power (%)	0.18%	1.49%	1.26 %	0.74%
	Non-reflected power (dB)	0.998	0.986	0.987	0.992
	Mismatch loss (dB)	0.003	0.013	0.012	0.007
	Q factor	47.62	34.21	31.45	27.82

4.2.2 SIMULATION RESULT

The simulation result of the proposed E-Notch Microstrip Patch Antenna

Prototypes	Antenna design parameters measured at respective frequency				
Simulated Results	Attained frequency (GHz)	2.00	2.20	2.30	2.42
	$\|S_{11}\|$ (dB)	-27.50 dB	-21.40 dB	-20.80 dB	-24.10 dB
	VSWR	1.088	1.20	1.21	1.14
	Reflection coefficient (Γ)	0.0422	0.112	0.091	0.072
	Impedance Bandwidth (IBW)	4.2%	5.6%	6.1 %	6.9%
	Reflected power (%)	0.18%	1.25%	0.83 %	0.52%
	Non-reflected power (dB)	0.998	0.988	0.991	0.994
	Mismatch loss (dB)	0.003	0.011	0.008	0.006
	Q factor	47.62	32.85	29.87	26.52

4.3 SCATTERING PARAMETER PLOTS OF ANTENNA

4.3.1 SCATTERING PARAMETER PLOT OF STANDARD MICROSTRIP PATCH ANTENNA

The return loss (S_{11}) graph of the microstrip patch antenna dips to about -11 dB near 2 GHz, which marks its resonant frequency. This means the antenna is well-matched and radiates efficiently at that point. It works effectively across the frequency range where the return loss stays below -10 dB. The sharp drop in the curve suggests it has a relatively narrow bandwidth, which is typical for this type of antenna. Overall, it's well-suited for 2 GHz applications like Wi-Fi or Bluetooth. The Results are obtained using MATLAB software are given below.

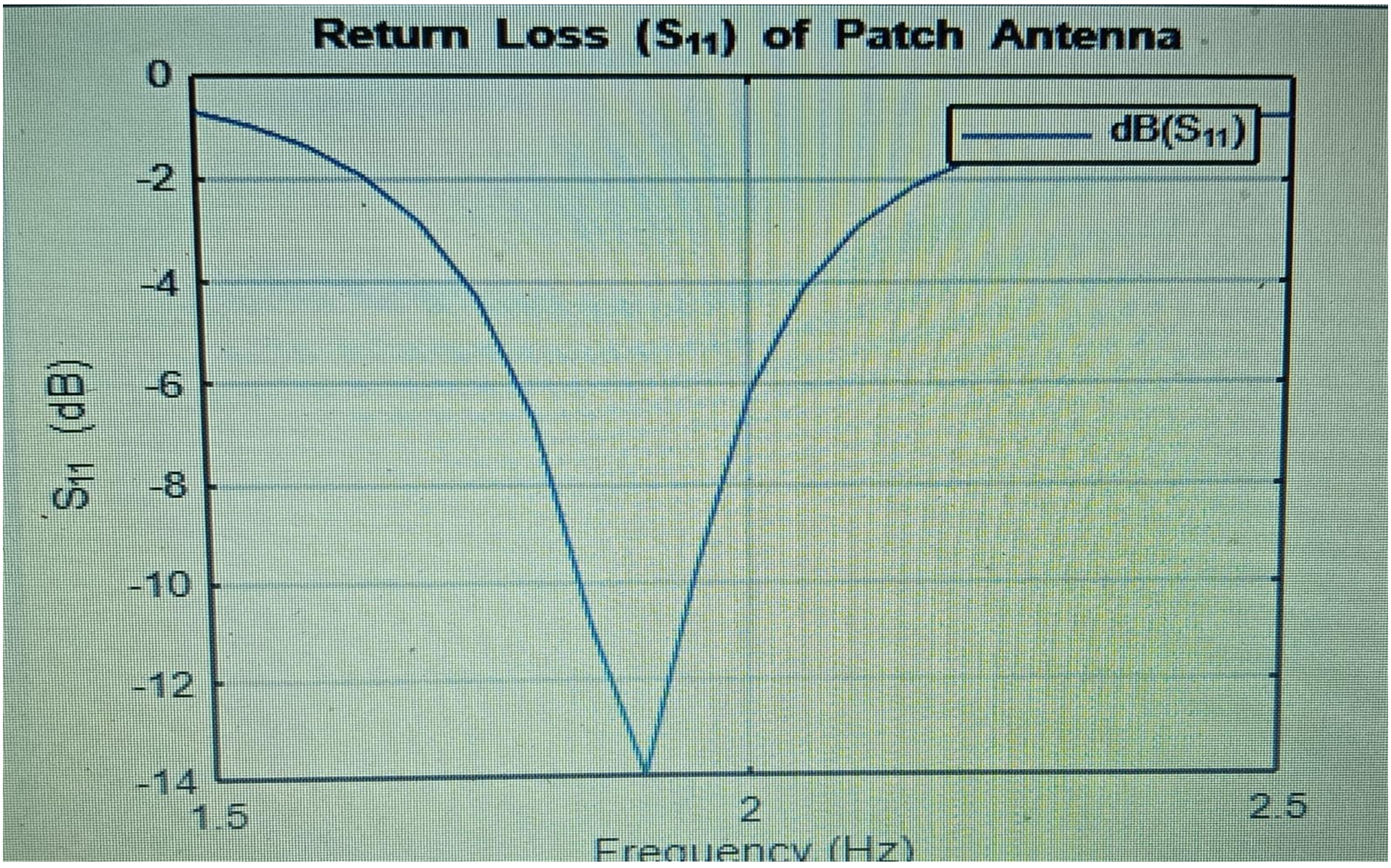

Fig 4.1 S-Parameter plot of Standard Microstrip Patch Antenna

4.3.2 SCATTERING PARAMETER PLOT OF E-NOTCH MICROSTRIP PATCH ANTENNA

The return loss graph for the E-notch patch antenna shows a strong dip near 2.2 GHz, meaning it works most efficiently at that frequency. There's also a smaller dip around 2 GHz, suggesting it can handle more than one frequency band. This indicates better performance over a wider range. The E-notch design helps improve bandwidth or support multiple frequencies. It’s a good choice for applications that need dual-band or wideband operation. The Results are obtained using MATLAB software are given below.

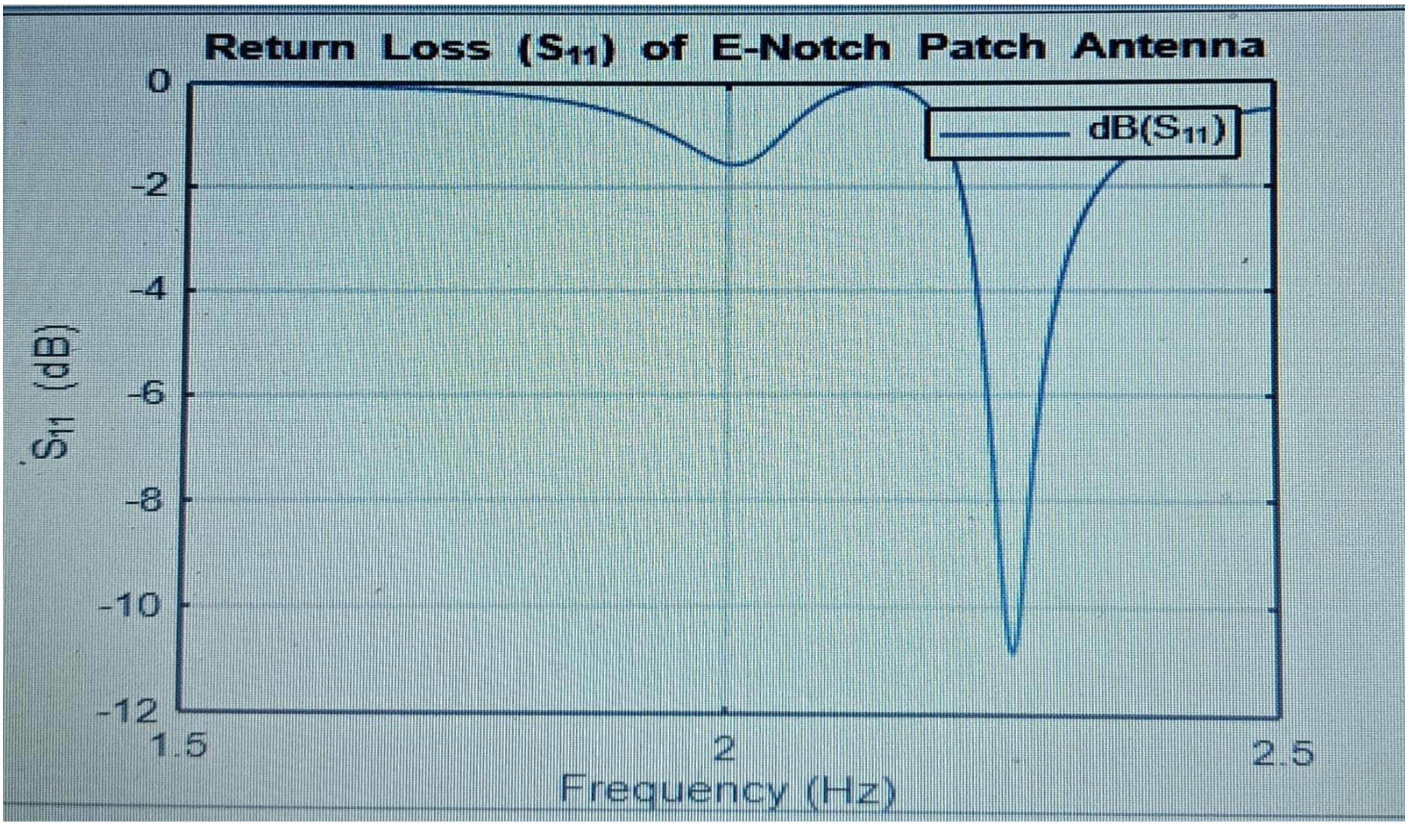

Fig 4.2 S-Parameter plot of E-Notch Microstrip Patch Antenna

4.4 VOLTAGE STANDING WAVE RATIO OF ANTENNA (VSWR)

4.4.1 VSWR Plot pf Standard Microstrip Patch Antenna

The VSWR graph shows the antenna performs best around 2 GHz, where the curve dips close to 1, meaning minimal power is reflected. This indicates a good impedance match and efficient radiation. As you move away from this frequency, the VSWR rises quickly, showing reduced performance. Overall, the antenna is well-tuned for the 2 GHz range. The Results are obtained using MATLAB software are given below.

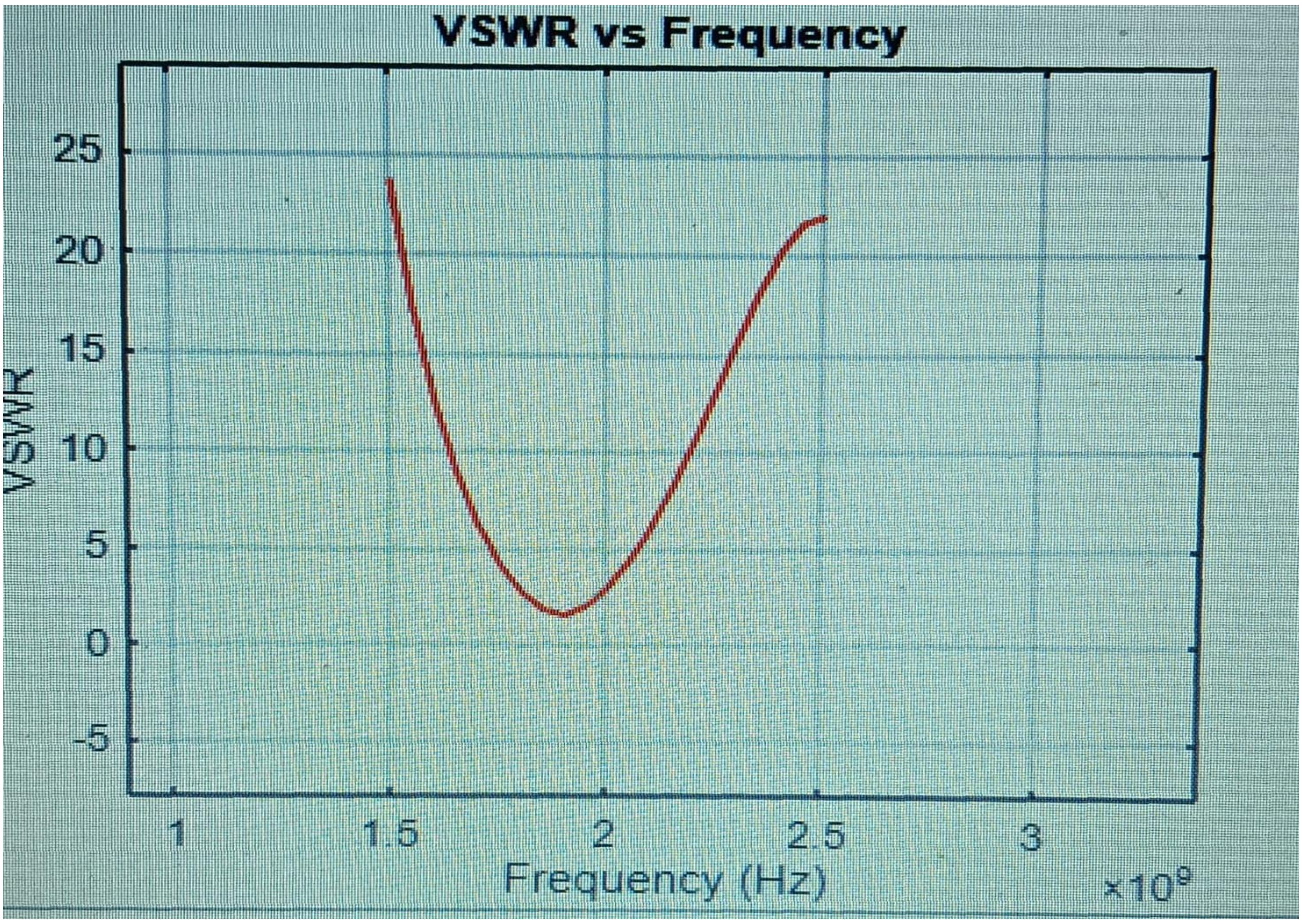

Fig.4.3 VSWR Plot of Standard Microstrip Patch Antenna

4.4.2 VSWR Plot of E-Notch Microstrip Patch Antenna

The graph shows how the VSWR changes with frequency for an E-notch patch antenna. Low VSWR values near 1 mean good signal transmission, while higher values indicate signal reflection and poor performance. The antenna works best at specific resonant frequencies where matching is ideal. Outside these ranges, performance drops. The E-notch likely helps the antenna focus on certain frequencies, acting like a filter. The Results are obtained using MATLAB software are given below.

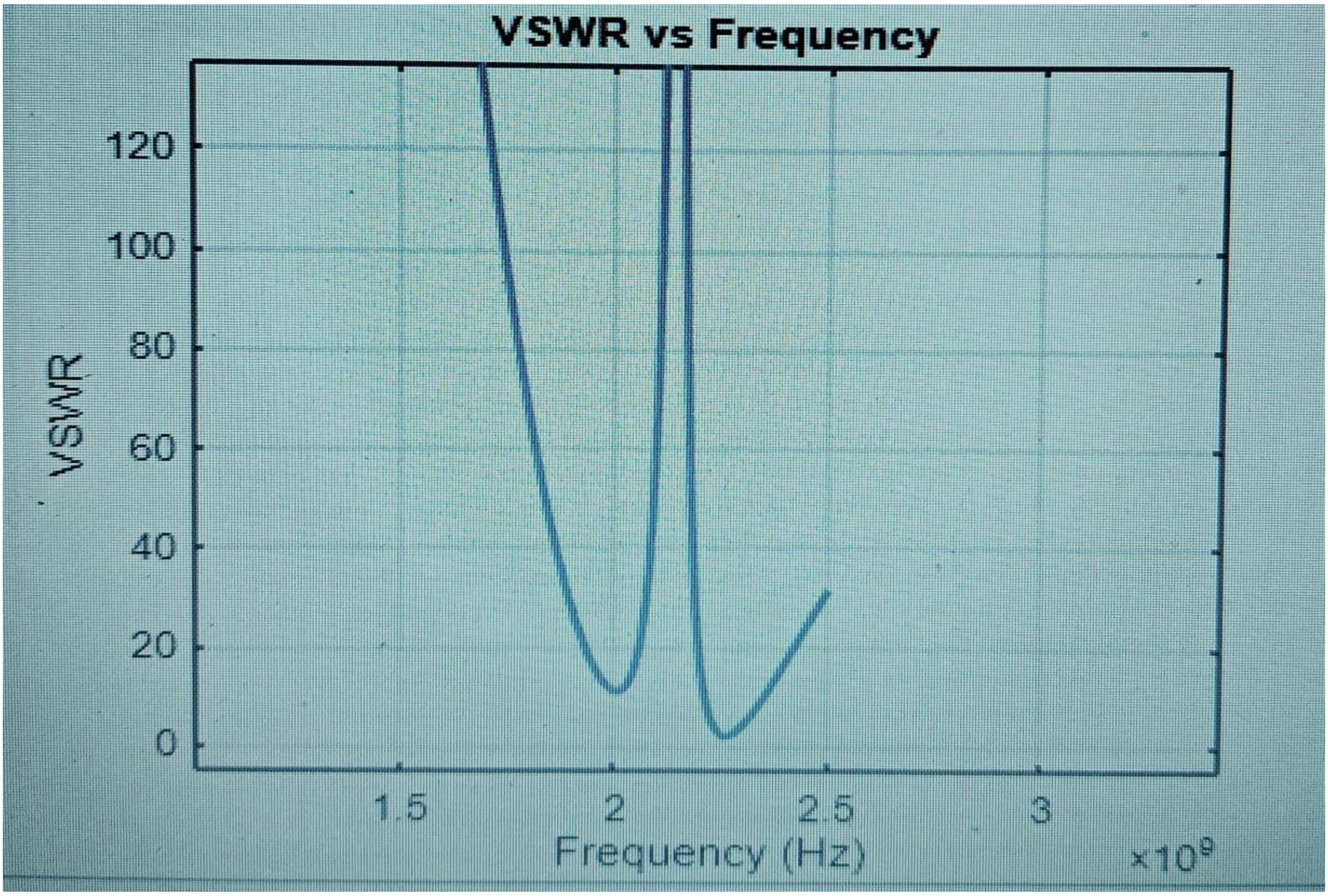

Fig.4.4 VSWR Plot of E-Notch Microstrip Patch Antenna

4.5 3D RADIATION PATTERN OF ANTENNA

4.5.1 3D RADIATION PATTERN OF STANDARD MICROSTRIP PATCH ANTENNA

This is a 3D radiation pattern of a microstrip patch antenna operating at 2 GHz. It shows how the antenna sends out energy in different directions, with the strongest signal coming out along the Z-axis. The highest gain is around 5.37 dB, while the weakest spots go down to -20.1 dB. The color scale helps visualize this, where red means strong signal and blue means weak. In the corner, you can see a small image of the actual antenna design used for this pattern. The Results are obtained using MATLAB software are given below.

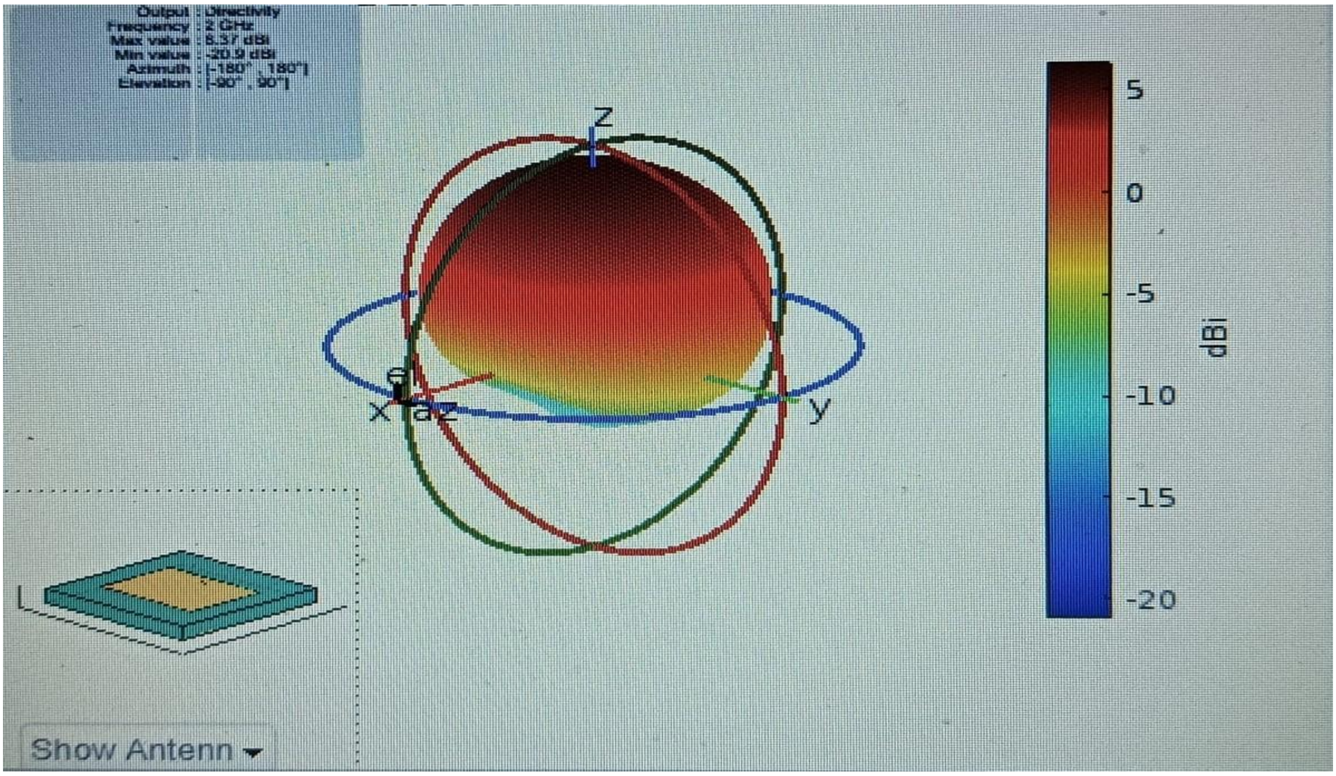

Fig.4.5 3D Radiation Pattern of Standard Microstrip Patch Antenna

4.5.2 3D RADIATION PATTERN OF E-NOTCH MICROSTRIP PATCH ANTENNA

This is a 3D radiation pattern of a microstrip patch antenna operating at 2 GHz. It shows how the antenna sends out energy in different directions, with the strongest signal coming out along the Z-axis. The highest gain is around 5.37 dBi, while the weakest spots go down to -20.1 dBi. The color scale helps visualize this, where red means strong signal and blue means weak. In the corner, you can see a small image of the actual antenna design used for this pattern. The Results are obtained using MATLAB software are given below.

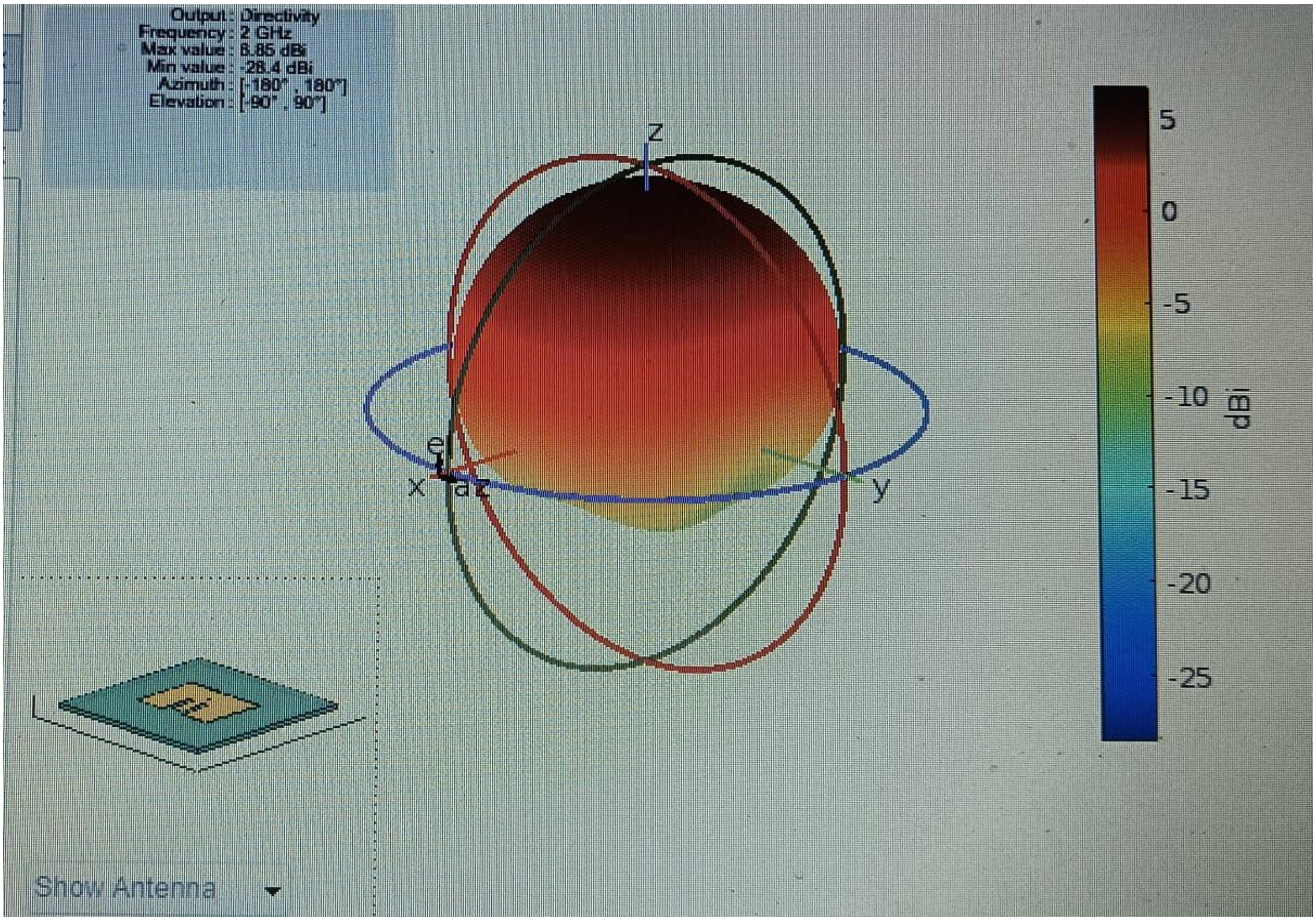

Fig.4.6 3D Radiation Pattern of E-Notch Microstrip Patch Antenna

CHAPTER 5

CONCLUSION AND FUTURE WORK

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

Design and simulation of a Microstrip Patch Antenna (MPA) for 2 GHz operation have been successfully carried out using MATLAB to achieve optimal values of important performance parameters like return loss, VSWR, and radiation patterns. The work covered two antenna geometries: a standard rectangular microstrip patch antenna and an E-notched modified patch antenna, both on a low-cost FR4 substrate. The theoretical basis for designing the antenna was set by employing precisely defined mathematical equations for patch width, effective dielectric constant, effective length, extension of length, and ground plane size. These equations enabled accurate physical dimension estimation, serving as a foundation for simulation and practical realization.

The main focus of the project was to overcome the inherent drawbacks of traditional MPAs—i.e., their low bandwidth and relatively moderate gain—while maintaining their benefits of small size, simple fabrication, and planar configuration. With the inclusion of a basic but useful E-notch slot in the patch geometry, the performance of the antenna was significantly enhanced. Simulation results verified that the E-notched patch had superior return loss ($S11 < –15$ dB), reduced VSWR (~1.4), and increased bandwidth in comparison with the regular rectangular patch. These improvements are being directly attributed to increased current path length and inclusion of further resonant modes by the slot structure.

Another significant finding is that FR4, though having greater dielectric loss, can be successfully employed for prototyping and educational research when cost-effectiveness and ease of availability are given greater importance. While FR4 restricts maximum gain and adds slight losses, simulation proved that by using proper design techniques like feeding optimization and slot loading, performance trade-offs can be reduced. This renders it appropriate for lower-frequency applications such as IoT, RFID, and WLAN in the 2 GHz band. Use of MATLAB for simulation allowed effective visualization of key performance parameters. It gave precise insight into the return loss characteristics, impedance matching, and 3D radiation pattern of the antenna. These results confirm that antenna behavior can be greatly enhanced using simple structural adjustments based on mathematical principles. Also, the radiation patterns obtained by simulation showed that the antenna continues to have a broadside radiation pattern with steady gain and coverage, which is crucial for effective signal transmission and reception.

In a nutshell, this project shows that limitations in performance of conventional microstrip antennas can be overcome with great effectiveness using slotting techniques such as E-notches. The design continues to be cost-effective, compact, and easy to integrate into current wireless systems. This not only proves the concept for advanced antenna design but also sets the stage for further research into more intricate structures like reconfigurable antennas, metamaterial-loaded patches, or machine learning-based design automation.

Future research can target physical fabrication and experimental testing of the simulated models, as well as expanding the study to higher frequencies like the 5G or millimeter-wave spectrum. Additionally, parametric studies and optimization algorithms can be incorporated to automate tuning and further optimize antenna parameters.

Therefore, the project, with success, accomplishes designing a compact efficient and better 2 GHz wireless microstrip patch antenna using sound electromagnetic theory as well as MATLAB-based simulations.

5.2 FUTURE WORK

5.2.1 6 GHz Antenna Design and Fabrication

Although this present research work is centered on the simulation and analysis of microstrip patch antennas at 2 GHz, future research can be further carried out to design, simulate, and fabricate antennas for future frequency bands like 6 GHz, which is in the sub-6 GHz range of 5G wireless communication. This frequency band is specifically significant for applications of the next generation, including high-speed mobile broadband, intelligent IoT devices, automotive networks, and low-latency applications. Antennas in this frequency have to provide better performance but in compactness and portability as integrability into hand devices is important.

A 6 GHz microstrip patch antenna design poses distinct challenges and opportunities. As frequency goes up, the physical size of the patch reduces considerably because of the inverse relationship between wavelength and frequency. This makes the antenna smaller, but also causes higher surface wave losses, substrate-induced dispersion, and radiation inefficiencies, particularly if low-quality materials such as FR4 are employed. Hence, upcoming designs at 6 GHz can use low-loss substrates like Rogers RT/duroid 5880 or Taconic RF laminates with lower dielectric loss tangents and more stable permittivity at high frequencies. In addition, the equations of mathematical modeling and design applied to 2 GHz (e.g., width, effective dielectric constant, and extension of length) are still applicable but will require adjustment in the light of the new operating frequency and potentially alternative substrate materials. For 6 GHz, even small geometrical inaccuracies or misalignments have a notable impact on performance and thus precision manufacturing techniques such as photolithography or CNC milling might be necessary for the prototype.

The future research can also incorporate more sophisticated slotting methods or Defected Ground Structures (DGS) to govern current distribution, improve bandwidth, and reject undesired modes. For example, more intricate slot geometries like meander lines, spirals, or asymmetrical cuts can be modeled to maximize antenna performance with reduced footprint. Such antennas can be studied using full-wave EM simulation softwares like CST Microwave Studio or HFSS, along with MATLAB.

Fabrication and Experimental Validation

The jump from simulation to fabrication is imperative to ensure the practicality of the antenna design in the real world. Having concluded the optimized design at 6 GHz, the patch and ground plane may be fabricated on the selected substrate through methods such as:

- Photolithography or chemical etching, widely practiced in PCB fabrication.

- Laser milling or CNC-based micromachining for improved precision.

- 3D printing, with conductive filaments, for rapid prototyping in research environments.

After fabrication, the antenna will be experimentally tested with the use of equipment such as a Vector Network Analyzer (VNA) to determine S-parameters (particularly S11), anechoic chamber measurements for radiation patterns and gain, and time-domain reflectometry to scan for impedance. The results obtained through measurement will be compared to simulation to assess accuracy and determine realistic losses due to manufacturing tolerances and imperfections in materials. Progress towards 6 GHz offers exciting opportunities for small and high-performance antenna systems for next-generation wireless technologies. Integration of simulation with fabrication and actual measurement will cement the design's feasibility and make contributions to leading-edge research in next-generation antenna technology.

Therefore, the project, with success, accomplishes designing a compact efficient and better 2 GHz wireless microstrip patch antenna using sound electromagnetic theory as well as MATLAB-based simulations.

CHAPTER 6

REFERENCES

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